

Vrajesh Sharma

Contact - 9429619257 Email - vrajesh.12042005@gmail.com Ahmedabad, Gujarat
linkedin.com/in/vrajesharma-7-dsa github.com/Vrajesh-Sharma vrajesharma.vercel.app

Summary

IEEE-published researcher and final year B.Tech student with research experience at IIT Patna in medical signal digitization. Skilled in end-to-end deep learning pipelines, LLM-based multi-agent systems, and RAG architectures, with experience building and deploying AI projects. Machine Learning Head at ASPDC, with experience teaching 150+ students through ML and DL programs.

Education

Adani University - Faculty of Engineering Science and Technology
B.Tech in Computer Science Engineering (AI & ML) — CGPA: 8.24

July 2023 – Expected May 2027
Ahmedabad, Gujarat

Experience

Research Intern, Indian Institute of Technology - Patna

Hybrid May 2026 – July 2026

- Designed and implemented an end-to-end CTG signal digitization pipeline to reconstruct fetal heart rate waveforms from challenging clinical monitor images using pixel-calibration methodology to map image coordinates directly to physiological measurements, improving robustness against glare, reflections, and incomplete grid structures.
- Engineered physiological signal features and evaluated classical machine learning and deep learning models for automated fetal state classification.

Machine Learning Head, ASPDC

Ahmedabad Aug 2025 – Aug 2026

- Leading a team of 5 Machine Learning Leads to design and deliver ML initiatives and learning programs for the club.
- Organized and taught the **ML in One Month** and **DL in One Month** programs for **150+** students, covering end-to-end ML pipelines, deep learning (ANN, CNN, RNN, LSTM), and model deployment workflows.

Machine Learning Lead, ASPDC

Ahmedabad Aug 2024 – Aug 2025

- Conducted an ML-101 workshop introducing Linear Regression and fundamentals of supervised learning.

Projects

TumorTrace — *PyTorch, U-Net, FastAPI, React, BraTS2020*

[Source Code](#) Mar 2026

- Built an end-to-end brain tumor segmentation system using a 2D U-Net trained on 57K+ MRI slices (45K train / 11K validation), achieving >0.87 Dice score on validation.
- Developed a full-stack ML application with FastAPI REST APIs and React, generating real-time tumor overlays, class-wise segmentation, and automated diagnostic reports with tumor area statistics.

LekhAI — *HTML, Flask, Gemini, DuckDuckGo, Render*

[Live Deployment](#) [Source Code](#) Nov 2025

- Built a multi-agent AI system for automated content generation using LLM orchestration and real-time web retrieval, reducing manual research and drafting effort by automating end-to-end blog generation.
- Designed a scalable agent architecture (Researcher, Strategist, Writer, Editor) supporting streaming responses, Markdown rendering, and structured multi-step reasoning workflows.

Sarathi — *Gemini, React, TypeScript, Flask*

[Live Deployment](#) [Source Code](#) Dec 2025

- Developed and deployed a RAG-based AI system leveraging vector search over curated datasets to deliver context-aware and citation-backed responses.
- Designed modular pipelines for retrieval, reasoning, and response generation, improving response relevance and grounding through knowledge-based augmentation.

Swabhav — *TensorFlow, Flask, Joblib, Scikit-learn, HTML*

[Live Deployment](#) [Source Code](#) Nov 2025

- Engineered and deployed a deep neural network-based personality classification system trained on 43K+ samples, achieving 89.06% test accuracy with strong generalization.
- Evaluated multiple models and selected optimal architecture based on accuracy, F1-score, and overfitting gap analysis.

Achievements

- First Author** — Published "*Empowering Text Classification with Agentic AI: A Systematic Review*" in IEEE Xplore, presented at the 2nd International IEEE Conference on Artificial Intelligence and Machine Vision. [View Paper](#)
- IEEE Member** and Volunteer at **IEEE AISYWLC 2025**.

Skills

Machine Learning & AI: Regression, Classification, Feature Engineering, Hyperparameter Tuning, Model Deployment

Deep Learning & GenAI: Neural networks, LLMs, RAG, Vector Databases, Multi-Agent Systems, Fine-tuning

Libraries & Frameworks: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, LangChain, CrewAI, Flask, FastAPI

Languages: Python, SQL, C, C++

Databases: MySQL, PostgreSQL, Pinecone, ChromaDB, Supabase

Tools & Platforms: Git, GitHub, Docker, REST APIs, VS Code, Vercel, Render, n8n